

U-VOWEL: Uncertainty-Routed VLA and Video-World-Model Policies for Out-of-Distribution Robot Manipulation

Anonymous Authors

Abstract—Generalist robot policies can follow diverse language instructions yet their reliability falls under changes in objects, scenes and task descriptions. This paper studies whether two action-generating policy families can compensate for different failures: a fast vision-language-action policy and a video-world-model policy that decodes actions from a video-trained predictive representation. We present U-VOWEL, a runtime framework that instruments each branch with native uncertainty evidence, maps recent evidence to a branch-specific failure score and routes control under a planning-cost budget. The controller uses the VLA as its fast path, invokes the video-world-model policy selectively and discards rejected proposals before any later fresh-state query. In the current single-seed evaluation on four selected LIBERO-PRO Goal perturbation suites, the selected controller completes 888 of 2,000 episodes, compared with 836 for the standalone VLA and 812 for the standalone video-world-model policy. Its mean wall time over successful episodes is 7.35 seconds, between the VLA’s 2.69 seconds and the video-world-model policy’s 14.16 seconds. The gains are concentrated in language and object perturbations, while swap and task perturbations remain difficult. These results motivate uncertainty-guided policy routing as a compute-aware approach to out-of-distribution manipulation and identify the failure regimes that still require new policy capability.

I. INTRODUCTION

Robot manipulation integrates perception, planning and control to modify the physical environment [1]. A deployed robot must preserve these capabilities as objects, workspaces and language instructions change. Learning from demonstrations provides a practical route to such behavior, yet training can cover only a limited subset of the states encountered during deployment. Small action errors may move the robot outside this subset and compound over a closed-loop rollout [2]. Generalization is therefore sequential: the robot must interpret an unfamiliar situation and keep its trajectory recoverable as every action changes the next observation.

Vision-language-action (VLA) models expand the range of situations that one policy can interpret. VLA architectures have progressed from language-conditioned transformers toward VLM-based policies and continuous action generation. Representative developments include RT-1, RT-2, OpenVLA, π_0 and OpenVLA-OFT [3]. Our fast branch follows SimVLA, a compact architecture that decouples perception from action generation [4]. These models condition direct action prediction on visual and language inputs. Their breadth does not establish robustness under every evaluation shift. LIBERO-PRO reports substantial degradation under controlled perturbations to object appearance, object position, semantic instructions, task definitions and environments [5].

This gap exposes a missing deployment capability. Producing an action does not tell the controller whether that action

is supported by familiar evidence or whether the policy is entering a failure regime. A generative action distribution may expose likelihood, variance or sample disagreement, but these quantities are not automatically calibrated probabilities of task failure. In VLAs, confidence calibration varies across action dimensions and over the task horizon [6], while per-step and per-degree-of-freedom entropy can contain brief failure-related spikes [7]. A robot therefore needs an operational way to recognize when its current proposal may be unreliable. Without this signal, it cannot decide when continued execution is justified, when a shorter replanning horizon is appropriate or when another policy should be consulted.

An uncertainty estimate becomes useful when it changes the control decision. A rejected proposal supplies no alternative behavior, whereas a second policy can propose actions through a different predictive process. Recent video-based robot policies show that predicted visual futures can support executable behavior [8], [9], [10]. This development motivates pairing a fast direct policy with a more expensive predictive policy.

This pairing creates a selective-control problem. A fixed policy choice cannot respond when reliability changes during an episode, whereas querying both policies at every decision incurs the cost of predictive generation even when the direct policy is reliable. The two branches also expose uncertainty evidence with different meanings and scales, which prevents a direct comparison of their raw signals.

We present U-VOWEL, an uncertainty-guided controller that uses the VLA as its fast default policy and selectively queries the video-world-model policy when the VLA proposal appears unreliable. Branch-specific temporal monitors convert each policy’s uncertainty evidence into routing decisions while limiting use of the more expensive predictive branch (Fig. 1).

The paper makes four contributions:

- (i) We introduce a heterogeneous closed-loop controller that routes between independently action-generating vision-language-action and video-world-model policies.
- (ii) We instrument both policies with uncertainty evidence. NLL-derived objectives train a heteroscedastic velocity-variance head for the VLA and a residual-variance head for video prediction, complemented by candidate disagreement and branch-native action or prediction statistics.
- (iii) We develop branch-specific temporal risk monitors that combine recent uncertainty evidence into failure scores and apply separately calibrated operating thresholds to the two policies.
- (iv) We introduce a selective-compute execution strategy

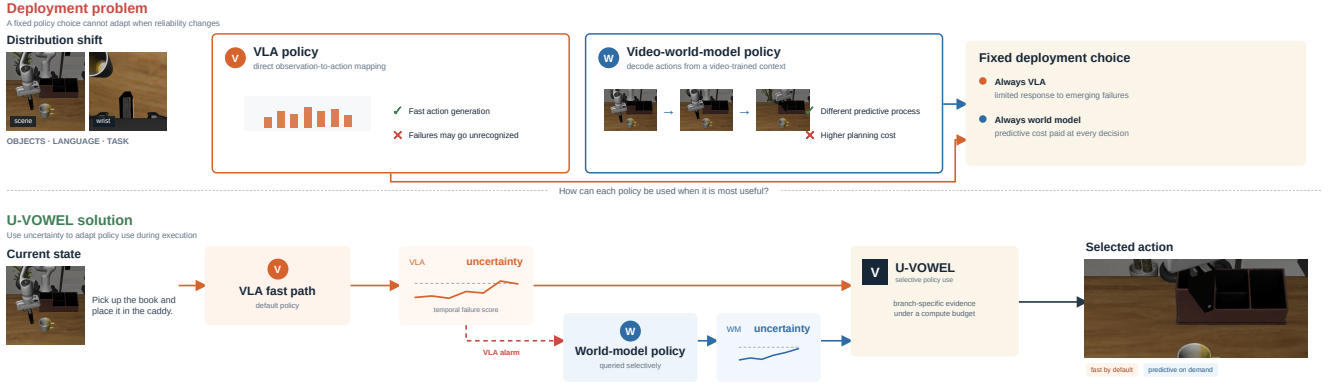


Fig. 1. Motivation for U-VOWEL. Under distribution shift, a direct VLA provides fast control but may not recognize emerging failures, while an always-on video-world-model policy incurs predictive planning cost at every decision. U-VOWEL uses branch-specific uncertainty to retain the VLA fast path and invoke the predictive policy selectively.

that keeps the VLA as the fast path, invokes the video-world-model policy after a VLA alarm and bounds repeated predictive queries through fresh-state escalation and a finite latch.

II. RELATED WORK

A. Manipulation learning and generalist policies

Imitation learning addresses sequential distribution mismatch primarily during training. DAgger collects expert labels on states induced by the learner and adds them to the training set [2]. This exposes the policy to recovery states that are absent from the original demonstrations, but requires interactive expert supervision. It does not provide a confidence signal for a frozen policy at deployment.

Action-sequence policies address a different aspect of closed-loop behavior: the temporal granularity of prediction and execution. ACT uses a conditional variational autoencoder to generate action chunks and temporally ensembles overlapping predictions [11]. Diffusion Policy represents multimodal action sequences through conditional denoising and executes them in a receding-horizon loop [12]. Chunking moves the unit of prediction from one action to a sequence and can smooth local behavior, while making the whole proposed sequence the relevant unit of reliability.

Generalist VLA policies place semantic task interpretation and continuous control within one learned pipeline [3]. Their episode outcomes consequently combine several capabilities, including visual grounding, instruction interpretation and action generation. LIBERO evaluates transfer across lifelong manipulation tasks, while LIBERO-PRO applies controlled distribution shifts to the benchmark [13], [5]. Task-level success measures the combined capability. Query-level analysis provides a different resolution by following how policy evidence evolves as successive action chunks alter the scene.

B. World models and action-generating video models

World-model policies introduce predicted consequences into action generation, but differ in how prediction is

connected to control. RoboDreamer treats a generated video as an intermediate plan and recovers actions through inverse dynamics [8]. Unified World Models instead couples action and future-observation diffusion, while Cosmos Policy jointly produces action chunks, future observations and predicted returns [9], [10]. The first design separates visual planning from action decoding; the latter designs place both outputs in one generative process.

This distinction determines where uncertainty can enter. A staged video policy can accumulate error in the predicted trajectory and in its action decoder, whereas a joint model links action quality to a shared generative representation. Both incur predictive computation that a direct VLA does not require. At the same time, their predicted futures expose intermediate structure that can be checked for consistency. Prior systems establish video-conditioned action generation as a viable policy family. Pairing such a policy with a separately trained direct policy additionally requires deciding when its predictive process is likely to be useful.

C. Uncertainty and runtime failure prediction

That decision depends on uncertainty evidence whose interpretation matches the generating process. Heteroscedastic likelihood learns input-dependent residual variance and represents conditional, or aleatoric, dispersion under the fitted model [14]. Gaussian NLL induces inverse-variance weighting that can progressively underweight poorly fitted regions; Beta-NLL modifies this weighting to reduce the imbalance [15]. Model uncertainty is commonly approximated through disagreement among independently trained predictors [16]. For large robot policies, however, maintaining several models adds training, memory and inference cost. Structured action outputs also make a single global confidence value difficult to interpret: VLA studies report that informative uncertainty varies across action dimensions and over time [7], [6].

A related line of work converts such evidence into temporal failure scores. FAIL-Detect learns a scalar score from successful trajectories and applies time-varying calibrated thresholds [17]. FIPER requires only successful rollouts: it aggregates

observation-embedding novelty from RND-OE and grid-based action-chunk entropy over short windows, calibrates both signals and raises an alarm through their logical conjunction [18]. SAFE instead learns failure prediction from temporal VLA features and studies transfer to held-out tasks [19]. Our VLA monitor follows the latter supervised setting but combines physical history, generated chunks, internal flow uncertainty and candidate disagreement in one learned risk score.

Calibration also determines what can be claimed about a detector. Conformal prediction provides finite-sample marginal coverage under exchangeability [20]; FIPER’s bound concerns alarms on successful in-distribution rollouts, not failure recall or coverage under arbitrary shift. Moreover, prior VLA detectors monitor one policy distribution. They do not establish that features or thresholds calibrated for a direct action model transfer to a predictive policy with different outputs and error mechanisms.

III. PROBLEM FORMULATION

We consider a language-conditioned manipulation episode $e \sim \mathcal{Q}$, where the evaluation distribution $\mathcal{Q}$ may differ from the training distribution through object appearance, scene configuration or task instruction. At control time t , the robot has visual observations $o_{0:t}$, proprioceptive states $q_{0:t}$, previously executed actions $a_{0:t-1}$ and an instruction ℓ . The evidence available to a causal controller is

$$E_t = (o_{0:t}, q_{0:t}, a_{0:t-1}, \ell). \quad (1)$$

The episode terminates at task completion or when its action budget is exhausted. Its outcome is $Y_e \in \{0, 1\}$, with $Y_e = 1$ denoting successful completion.

The system contains two frozen action-generating policies indexed by $b \in \{V, W\}$. The VLA policy V maps the current evidence directly to an action chunk. The video-world-model policy W processes the evidence with a video-prediction backbone and decodes an action chunk from its intermediate predictive context. Querying branch b returns

$$(A_t^b, u_t^b) = \pi_b(E_t), \quad A_t^b \in \mathbb{R}^{H_b \times d_a}, \quad u_t^b \in \mathbb{R}^{d_b}, \quad (2)$$

where H_b is the proposed horizon, d_a is the action dimension and u_t^b contains the uncertainty evidence exposed by that branch. The two uncertainty vectors need not share a dimension, scale or probabilistic interpretation.

At each decision point, the controller queries V and may additionally query W . Let $n_t^b \in \{0, 1\}$ indicate whether branch b is queried, with $n_t^V = 1$. From the proposals available at time t , a routing rule ρ selects a branch b_t and an executed prefix length $m_t \in \{1, \dots, H_{b_t}\}$. The applied controls are $A_t^{b_t}[1 : m_t]$. Both the selected chunk and the evidence used to select it must correspond to the same environment state E_t ; once an action changes the state, an unexecuted proposal cannot be treated as a current decision.

Let c_b be the measured cost of querying branch b , with the predictive branch generally satisfying $c_W > c_V$. The planning

cost of episode e is

$$C_e(\rho) = \sum_t \sum_{b \in \{V, W\}} c_b n_t^b. \quad (3)$$

The desired router maximizes task success on the evaluation distribution while respecting an expected planning budget B :

$$\max_{\rho} \mathbb{E}_{e \sim \mathcal{Q}}[Y_e] \quad \text{subject to} \quad \mathbb{E}_{e \sim \mathcal{Q}}[C_e(\rho)] \leq B. \quad (4)$$

Problem. Given the frozen policies π_V and π_W , their branch-native uncertainty evidence and calibration trajectories, construct a causal routing rule ρ that selects executable action prefixes at query time and improves out-of-distribution task success without exceeding the planning budget.

IV. METHODOLOGY

A. System overview

U-VOWEL connects two independently trained action-generating policies through the interface in Eq. (2). Both branches receive the same language instruction from a synchronized decision state, but retain their native observations, action representations and generative processes. The direct VLA forms the default path. The video-world-model policy is queried only when the VLA monitor raises an alarm. Figure 2 follows one such decision from observation to execution.

Each policy first produces an action proposal and uncertainty measurements tied to its own prediction process. A branch-specific temporal model converts those measurements into a failure score. The controller then accepts one proposal, shortens VLA execution or temporarily suppresses repeated calls to the predictive branch. This separation prevents predicted variance and sample disagreement from being treated as calibrated probabilities of task failure.

B. Direct VLA policy

1) *Action generation:* The direct branch modifies SimVLA [4]. A frozen SmolVLM-500M encoder processes the language instruction and current agent and wrist RGB views. A flow-matching action transformer combines this representation with an eight-dimensional proprioceptive state and generates a normalized chunk of $H = 10$ seven-dimensional Cartesian actions. For an expert chunk A^* , Gaussian noise ϵ and flow time $\gamma = 0.001 + 0.999\tilde{\gamma}$, with $\tilde{\gamma} \sim \text{Beta}(1.5, 1)$, training constructs

$$X_{\gamma} = (1 - \gamma)A^* + \gamma\epsilon, \quad V_{\gamma}^* = \epsilon - A^*. \quad (5)$$

The action transformer predicts the velocity field from (X_{γ}, γ) , then inference integrates that field from noise to an action chunk. The vision-language encoder remains frozen during uncertainty fine-tuning; the action transformer and its output heads are trainable.

2) *Heteroscedastic velocity uncertainty:* We replace the original velocity output with parallel velocity and variance heads. For action index i , the second head emits $r_{\theta, i}$, which is mapped to a positive conditional variance,

$$\sigma_{\theta, i}^2 = \text{softplus}(r_{\theta, i}) + \varepsilon_n, \quad \varepsilon_n = 10^{-6}. \quad (6)$$

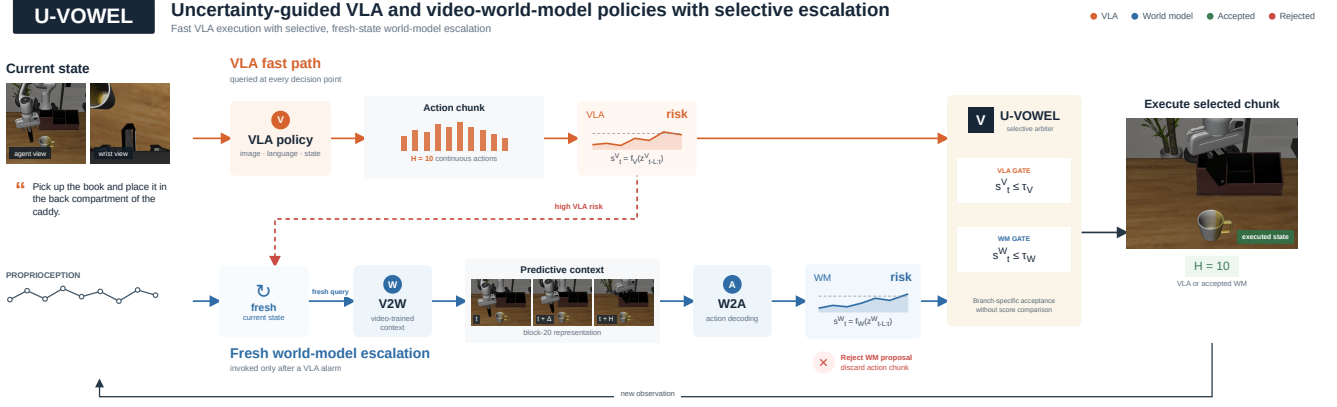


Fig. 2. U-VOWEL execution loop. The VLA supplies the default action chunk and a branch-specific failure score. A VLA alarm makes the predictive branch compute a fresh video-conditioned context and decode an alternative action chunk from the current state. The visual frames represent the V2W training target; the deployed branch passes an intermediate representation to W2A without decoding a pixel-space rollout. The two scores are evaluated against separate thresholds and are never compared numerically.

The modified policy is fine-tuned with an element-wise Beta-NLL objective [15]:

$$\mathcal{L}_V = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \text{sg} \left[(\sigma_{\theta,i}^2)^\beta \right] \times \left[\frac{(V_{\theta,i} - V_{\gamma,i}^*)^2}{2\sigma_{\theta,i}^2} + \frac{1}{2} \log \sigma_{\theta,i}^2 + \kappa \right], \quad \beta = 0.5. \quad (7)$$

Here sg stops gradients through the Beta-NLL weight, $\mathcal{I}$ spans action timesteps and dimensions and $\kappa = -\frac{1}{2} \log \varepsilon_n$. This constant keeps the bracketed term nonnegative for $\sigma_{\theta,i}^2 \geq \varepsilon_n$. The residual term weights velocity error by predicted precision, while the logarithmic term penalizes indiscriminate variance inflation. The detached Beta-NLL factor reduces the attenuation of high-variance examples without adding another gradient path through the variance weight.

At inference, variance is retained along the Euler trajectory. A diagonal first-order approximation accumulates

$$P_{\gamma+\Delta\gamma} = P_\gamma + (\Delta\gamma)^2 \sigma_\theta^2(X_\gamma, \gamma). \quad (8)$$

Each query draws nine independently seeded $H = 10$ chunks from one encoded observation. Candidate zero is the main proposal; eight alternatives provide uncertainty evidence. A 49-dimensional record combines the main candidate's last-step and path-integrated variance, denoising dynamics, chunk geometry and dispersion across all nine samples. Separately, a seven-dimensional candidate-disagreement descriptor is computed from the eight normalized alternatives: centered and pairwise chunk distances, overall dispersion and translation, rotation, gripper and flattened-coordinate dispersion. This descriptor is not FIPER's entropy-based ACE. The deployed monitor retains eight of the 49 dimensions through a ranking performed on seen train/validation data, covering initial denoising magnitude, update-direction flips, path variance and action-sample dispersion. None of these inputs contains a task identifier, reward or future observation.

C. Predictive video-action policy

1) *V2W-W2A generation*: The predictive branch separates a video-trained representation from action decoding. Its video-to-world component (V2W) adapts a Cosmos Predict2 backbone through rank-256 LoRA and trainable adaptive-normalization parameters [21], [10]. V2W is trained to predict future video from five 480×640 agent-view frames sampled from recent observation history and the language embedding. The deployed policy does not complete or decode that visual rollout. It stops at the first V2W denoising evaluation and extracts a hidden representation after transformer block 20 as the predictive context C_t . The world-to-action component (W2A) is a separate diffusion transformer trained while V2W remains frozen. It conditions on C_t , the ten-dimensional robot state and the video noise level, then generates 60 ten-dimensional tokens. Translation, six-dimensional rotation and gripper outputs are decoded to the seven-dimensional environment action.

W2A uses the interpolation and velocity target in Eq. (5), with $\gamma = 0.001 + 0.999\tilde{\gamma}$ and $\tilde{\gamma} \sim \text{Beta}(1, 1)$, and minimizes

$$\mathcal{L}_{W2A} = 10 \mathbb{E}_{\gamma, \epsilon} \left[\|V_\omega(X_\gamma, \gamma; q_t, C_t, \sigma) - V_\gamma^*\|_2^2 \right], \quad (9)$$

where σ is the V2W noise level. During training, the observation condition is dropped with probability 0.2. This follows the flow-matching formulation [22]. Although W2A predicts a long sequence, the controller executes at most ten actions before observing the environment again.

Repeated W2A samples expose action uncertainty without repeating the expensive video pass. In the K1 configuration, one V2W context produces eight W2A candidates. Their per-step and per-dimension variance, pairwise mean-squared distance, receding-plan disagreement and denoising dynamics form the K1 uncertainty vector. Candidate zero remains the executable proposal; the other samples provide evidence and do not compete for control. K3 instead generates three V2W contexts. The main context produces eight W2A candidates

and each additional context produces one fixed-seed candidate, giving ten action chunks for measuring sensitivity to the sampled V2W representation. Thus K1 and K3 name the number of V2W contexts, not the number of executed chunks or the monitor history.

2) *Residual uncertainty in V2W*: Separately, we train a lightweight residual-variance head on the frozen V2W backbone. Let $h_{20,i}$ be a future token at block 20 and let ρ_i^2 be the mean squared clean-latent residual over its 16 channels. A calibration table supplies a positive base variance $b_{q(\sigma)}$ for the current video-noise bin. The head predicts a token-wise correction from $h_{20,i}$ and an embedding of $\log \sigma$:

$$\begin{aligned} \ell_i &= \text{clip}(\log b_{q(\sigma)} + \Delta_\phi(h_{20,i}, \log \sigma), -8, 5), \\ v_i &= \exp(\ell_i). \end{aligned} \quad (10)$$

Only the variance head and noise embedding are optimized. With $w_i = 1$ for future tokens and zero for observed conditioning tokens, the loss is

$$\mathcal{L}_{\text{V2W-NLL}} = \frac{1}{2} \frac{\sum_i w_i (\rho_i^2 / v_i + \ell_i)}{\sum_i w_i + \varepsilon_n}. \quad (11)$$

This head estimates conditional latent-residual scale. It does not by itself measure task failure or complete epistemic uncertainty. Its summaries enter the uncertainty-feature ablations. The selected K1 controller uses the action-uncertainty profile described above and therefore does not depend on V2W variance-head features.

D. Branch-specific temporal risk

Let R_t^b denote the normalized query record built for branch b . The record may contain sequential action or uncertainty tokens, a bounded history and current summary features. Each monitor computes

$$s_t^b = \text{sigmoid}(f_b(R_t^b)), \quad b \in \{V, W\}. \quad (12)$$

The records and functions are branch-specific because the two generative models expose different evidence.

1) *VLA temporal risk monitor*: The VLA monitor takes FIPER [18] as a methodological starting point. FIPER accumulates RND-OE observation-embedding novelty and grid-based action-chunk entropy (ACE) over short windows, calibrates both signals from successful rollouts and combines their alarms by a logical AND. Our VLA monitor instead learns one supervised score from seen successful and failed rollouts, using candidate-action evidence and its recent history. Its seven-value candidate-disagreement descriptor is not FIPER ACE, and it uses neither RND nor direct observation embeddings.

The VLA monitor uses a late-fusion Transformer. Inputs are standardized using statistics fitted only on the training split. Its sequence path contains 16 history tokens of dimension 21—proprioception (8), the previous executed action (7) and six candidate-disagreement values—followed by ten 7D tokens from the normalized main proposal. Separate linear projections map both token types to width 128. A learned CLS token and positional embeddings precede a three-layer, four-head Transformer with 512-dimensional feed-forward blocks.

Startup histories are zero-padded and processed without an attention mask.

The static 51D path concatenates 28 main-chunk statistics (first, mean, standard deviation and endpoint change), the seven candidate-disagreement values, current 8D proprioception and the selected eight uncertainty features. A linear-GELU projection maps it to width 128. Its output is concatenated with the Transformer CLS output, normalized and passed through a 128D hidden layer to the risk logit. Neither task identity nor timestep is an input.

The VLA monitor is trained as a binary classifier from rollout outcomes. A query in a failed rollout receives a failure target even when that query did not cause the eventual failure. Let $F_e = 1 - Y_e$ denote episode failure, and let N_0 and N_1 be the numbers of successful and failed training rows. Its positive-class weight is $p = \max(1, N_0 / \max(1, N_1))$, and the implemented objective is

$$\mathcal{L}_{\text{risk}}^V = -\frac{1}{N} \sum_{e,t} [p F_e \log s_{e,t}^V + (1 - F_e) \log(1 - s_{e,t}^V)]. \quad (13)$$

Consequently, s_t^V is an empirical warning score for the observed episode prefix, not a causal probability that executing A_t^V will fail.

VLA operating points are computed on a held-out seen validation split: one maximizes row-level F1 using both classes, while success-only alternatives use empirical quantiles of successful row scores. The operating point is selected according to the desired seen false-alarm/detection trade-off and frozen before unseen evaluation. An episode alarms if any query crosses it. This is empirical row-score calibration, not an episode-level conformal guarantee.

2) *World-model temporal risk monitor*: The selected K1 world-model monitor uses an eight-query history and a two-layer GRU. At each query it receives 74 scalar W2A uncertainty features, including candidate variance, pairwise disagreement, plan overlap and denoising summaries. Sixteen temporal action-uncertainty tokens preserve the per-step structure of those statistics. The GRU and static branch have width 128 and their fused representation passes through a 64-dimensional latent layer to the risk logit. V2W residuals, multi-context disagreement and candidate centrality define richer profiles used in the ablations, but are absent from this K1 feature contract.

The world-model threshold is calibrated from the episode maximum score on successful held-out rollouts. For n calibration episodes with maxima $g_e = \max_t s_{e,t}^W$, its level- α threshold is the corrected empirical quantile

$$\tau_W(\alpha) = g_{(\min\{n, \lceil (n+1)(1-\alpha) \rceil\})}, \quad (14)$$

where $g_{(k)}$ is the k -th order statistic. This controls the marginal episode-level false-alarm rate under the usual exchangeability assumption [20]; it does not guarantee calibration after arbitrary sequential distribution shift.

E. Selective execution

Algorithm 1 combines the two monitors without comparing their score magnitudes. The state variable c is the number of

Algorithm 1 U-VOWEL selective execution for one episode

Require: policies π_V, π_W , monitors f_V, f_W , thresholds τ_V, τ_W , horizons $H > h$, latch span C

```
1:  $c \leftarrow 0$  ▷ remaining ineligible environment actions
2: while episode active and action budget remains do
3:   acquire current evidence  $E_t$ 
4:    $(A^V, u^V) \leftarrow \pi_V(E_t)$ ;  $s^V \leftarrow f_V(R_t^V)$ 
5:   if  $s^V \leq \tau_V$  then
6:      $m \leftarrow H$  if  $c = 0$ , else  $\min(H, c)$ 
7:      $n \leftarrow \text{EXECUTEPREFIX}(A^V, m)$ ;  $c \leftarrow \max(c - n, 0)$ 
8:   else if  $c > 0$  then
9:      $n \leftarrow \text{EXECUTEPREFIX}(A^V, \min(h, c))$ ;  $c \leftarrow \max(c - n, 0)$ 
10:  else
11:     $(A^W, u^W) \leftarrow \pi_W(E_t)$  ▷ fresh query from the current state
12:     $s^W \leftarrow f_W(R_t^W)$ 
13:    if  $s^W \leq \tau_W$  then
14:       $\text{EXECUTEPREFIX}(A^W, H)$ 
15:    else
16:       $\text{DISCARDPROPOSAL}(A^W)$ 
17:       $c \leftarrow C$ 
18:       $n \leftarrow \text{EXECUTEPREFIX}(A^V, \min(h, c))$ ;  $c \leftarrow \max(c - n, 0)$ 
19:    end if
20:  end if
21: end while
```

environment actions remaining before the world-model branch becomes eligible after a rejected proposal. When $c = 0$, a VLA alarm triggers a fresh V2W-W2A query from the same evidence E_t . An accepted world-model proposal supplies the next action prefix. If its own monitor also alarms, the proposal is discarded, the controller executes a shorter prefix of the already available VLA chunk and starts the latch. The selected K1 controller uses $H = 10$, $h = 5$ and $C = 50$.

While the latch is active, the VLA is still queried and scored at every decision point. A high VLA score no longer invokes V2W-W2A and receives the short horizon h . A low score may use the ordinary horizon H , capped at latch expiry so that eligibility is reconsidered at the next state. The latch is decremented by actions actually applied, including early termination within a chunk. Once it expires, the next VLA alarm runs new world-model inference. The rejected action buffer is never reused; its score remains only in the monitor history and audit log.

V. EXPERIMENTAL PROTOCOL

A. Evaluation suites

- **LIBERO in-distribution evaluation:** standard LIBERO suites will measure task competence before perturbation. Unperturbed LIBERO-Goal episodes will also provide the in-distribution reference for the corresponding LIBERO-PRO tasks.
- **LIBERO-PRO evaluation:** the official Goal-Language, Goal-Object, Goal-Swap and Goal-Task suites contain ten tasks with 50 initial states per task. Their matched evaluation comprises 2,000 episode identities per policy: four suites, ten tasks per suite and 50 initial states per task.
- **Project-specific Goal-Object-OOD evaluation:** a separate 18-task suite contributes 50 initial states per task, for 900 matched episode identities per policy. We report

it separately and do not present it as an official LIBERO-PRO suite.

- **IsaacLab evaluation:** the Franka environment will separate in-distribution trials from controlled changes in object identity, category, geometry, material, pose, lighting, background, camera and combinations of these factors. Asset-family-disjoint splits will support the unknown-object analysis.
- **Physical evaluation:** frozen simulation-selected checkpoints, monitors and thresholds will be transferred to the real Franka setup. Trial blocks will cover familiar objects, novel identities within known categories and held-out object categories.

Within each completed LIBERO protocol, an episode identity is the tuple of suite, task, initial-state index and evaluation seed. Compared policies share the complete identity set and instruction, and the campaigns use one evaluation seed with 50 initial states per task. Observations are rendered at 480×640 and passed through each model’s configured preprocessing. The action budget is 250 environment actions; success is the environment task- completion signal and exhausting the budget without completion is a failure. Normal execution uses ten-action prefixes. The selected controller uses $H = 10$, a five-action VLA prefix after mutual rejection and a 50-action world-model eligibility latch. Branch thresholds are calibrated independently before online evaluation and remain frozen on the matched test episodes.

B. Baselines and comparison policy

- The matched LIBERO comparison comprises standalone HF SimVLA H10, standalone base world-model H10, fresh re-query U-VOWEL and the selected K1 latch-50 U-VOWEL controller.
- Historical cached K1/K3 controllers and the CUDA legacy-cache reproduction are retained only as diagnostic ablations. Because they may retain rejected world-model state, they are excluded from the headline controller claim.
- Branch-specific ablations compare base and uncertainty-instrumented policies, monitor architectures and K1/K3 predictive-context variants under their corresponding matched manifests.
- Reproduced state-of-the-art policy baselines will use the same observations, prompts, action interface and episode manifests whenever their implementations permit. Published results obtained under different protocols will be reported in a separate comparison block and excluded from matched significance tests.
- The IsaacLab environment is project-specific. Its comparison will use strong policy families adapted to the same data and control interface rather than presenting an unmatched public-leaderboard claim.

C. Metrics and statistical analysis

- Primary task metrics are episode success, per-task success and suite-level success. Paired outcome tables

will count shared successes, shared failures and one-policy-only successes on identical episodes.

- Offline monitor metrics include AUROC, AUPRC, risk-coverage behavior, failure detection, success false alarms, detection time and warning lead time where the labels support it. A success false alarm is any alarm in a successful episode; failure detection is the fraction of failed episodes containing an alarm. Det@10, Det@25 and Det@50 measure failures first detected within 10%, 25% and 50% of the evaluated horizon. Never is the fraction of failed episodes with no alarm.
- Controller metrics include branch-query counts, escalation rate, world-model acceptance, mutual rejection, useful and harmful outcome reversals and latch activity.
- Efficiency metrics include branch inference time, monitor overhead, all-episode wall time, successful-episode wall time, throughput and peak GPU memory. Success-conditioned time will always be reported beside the all-episode statistic. Episode timing is runner wall time and excludes one-time model loading.
- The completed LIBERO campaigns contain one evaluation seed and repeated initial states nested within tasks. Exact McNemar tests describe paired episode differences but do not account for task clustering or model-selection history. We therefore report task-level results and do not treat episodes as independent task samples.

D. Sim-to-real protocol and safety

- Simulation and hardware interfaces will be checked for camera geometry, color conversion, state representation, control frequency, workspace frame and action scaling.
- Policy order will be randomized within object and condition blocks. The VLA, world-model and fused controller will use the same trial manifest. Checkpoints, prompts, thresholds and stopping rules will remain frozen during the reported evaluation.
- Outcomes will distinguish success, drop, collision, time-out, automatic stop and human intervention. Independent workspace, velocity, force, watchdog and emergency-stop limits remain outside the learned controller.

VI. RESULTS

A. Seen-calibrated VLA risk under OOD transfer

We first isolate the VLA monitor from the online controller. The promoted SimVLA checkpoint was selected by seen-validation AUPRC (0.9369; AUROC 0.9345), and every operating point was fitted on held-out successful or labeled seen rollouts. We then froze the checkpoint and thresholds and pooled six unseen suites containing 680 episodes: 331 successes and 349 failures. The official FIPER reproduction uses the same OOD episodes, five RND seeds and its three calibration rules, with calibration restricted to 150 successful seen Goal-Object episodes.

Figure 3 shows the resulting episode-level trade-off. At the seen-success $q_{0.95}$ threshold, our temporal monitor detects 92.55% of failures with 39.27% success false alarms. Moving to $q_{0.99}$ reduces false alarms to 3.63%, but also

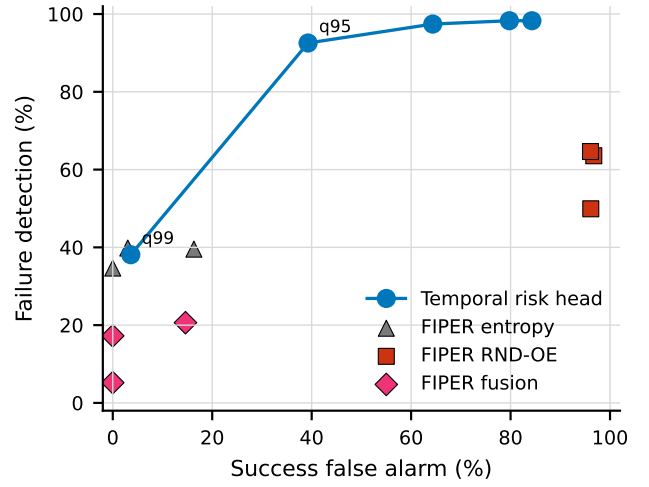


Fig. 3. Seen-calibrated monitor transfer to six unseen suites. Each point is computed on the same 331 successful and 349 failed episodes. The temporal risk-head curve varies only thresholds fixed on seen data; the three points per FIPER family are its CT-quantile, TVT-quantile and TVT-CP-band rules. Neither method is recalibrated on the OOD outcomes.

reduces detection to 38.11%. FIPER entropy occupies a similar low-false-alarm regime (0–16.31% false alarms and 34.67–39.83% detection). RND-OE reaches at most 64.64% detection while alarming on at least 96.07% of successful episodes; consequently, the logical-AND fusion suppresses both signals and detects only 5.21–20.63% of failures. Thus, no detector dominates at every operating point. The learned temporal monitor contributes a high-recall region that the seen-calibrated FIPER rules do not reach, while FIPER entropy remains competitive when very low false-alarm operation is required.

B. Matched LIBERO-PRO evaluation

Table I and Fig. 4 report 2,000 matched episodes per policy. U-VOWEL with the explicit latch-50 controller reaches 888/2,000 successes (44.40%), compared with 836/2,000 (41.80%) for SimVLA and 812/2,000 (40.60%) for the world model. The absolute gains are 52 and 76 episodes, respectively. They are concentrated in Goal-Language and Goal-Object. All policies remain weak on Goal-Swap and Goal-Task, where routing cannot recover a capability absent from both branches.

The paired outcomes in Table II confirm that the branches are complementary rather than nested. Relative to SimVLA, U-VOWEL gains 149 episodes and loses 97 (exact McNemar $p = 0.00110$); relative to the world model, it gains 207 and loses 131 ($p = 0.00004$). Fresh re-query and latch-50 are statistically indistinguishable on this manifest ($p = 0.79137$), which supports selecting the latch by compute rather than by a success claim. These tests are descriptive: the evaluation uses one seed and repeated initial states nested within tasks.

On the separate project-specific Goal-Object-OOD protocol, the same ranking is observed over 900 matched episodes: 823 successes for the world model (91.44%), 841 for SimVLA (93.44%), 859 for fresh re-query (95.44%) and 861 for latch-

TABLE I

MATCHED OFFICIAL LIBERO-PRO SUCCESS. EACH SUITE CONTAINS 500 EPISODES (10 TASKS, 50 INITIAL STATES PER TASK); OVERALL CONTAINS 2,000 EPISODES. BOLD MARKS THE BEST RESULT IN EACH COLUMN.

Policy	Goal-Language	Goal-Object	Goal-Swap	Goal-Task	Overall
World model H10	387 (77.4%)	352 (70.4%)	25 (5.0%)	48 (9.6%)	812 (40.60%)
HF SimVLA H10	397 (79.4%)	406 (81.2%)	20 (4.0%)	13 (2.6%)	836 (41.80%)
Fresh U-VOWEL	410 (82.0%)	415 (83.0%)	18 (3.6%)	42 (8.4%)	885 (44.25%)
Latch-50 U-VOWEL	408 (81.6%)	423 (84.6%)	16 (3.2%)	41 (8.2%)	888 (44.40%)

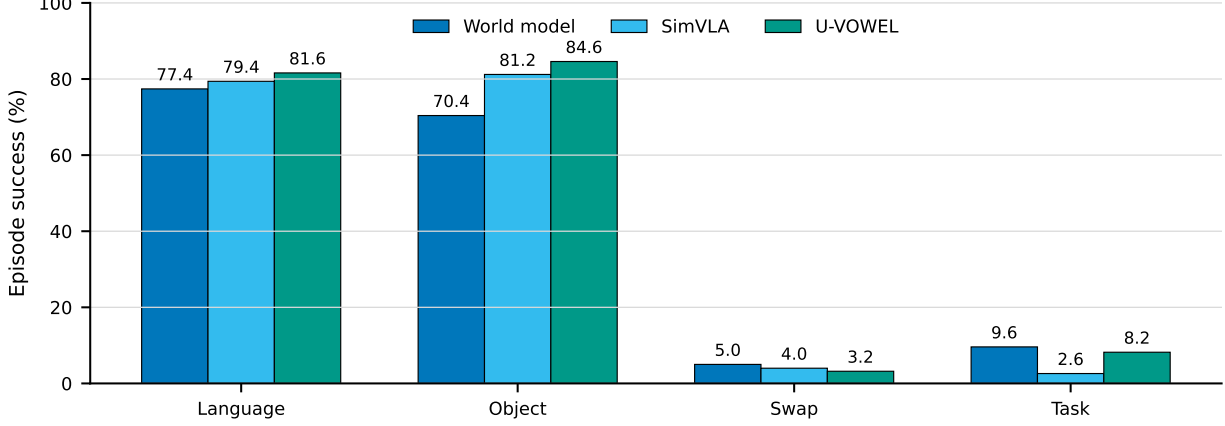


Fig. 4. Suite-level success on matched official LIBERO-PRO episodes. The selected U-VOWEL controller improves the aggregate result through the Language and Object suites, but does not solve the Swap or Task shifts.

TABLE II

PAIRED OFFICIAL LIBERO-PRO OUTCOMES FOR LATCH-50 U-VOWEL.

Comparator	Both	U only	C only	Neither	p
World model	681	207	131	981	0.00004
SimVLA	739	149	97	1,015	0.00110
Fresh U-VOWEL	858	30	27	1,085	0.79137

50 U-VOWEL (95.67%). We keep this result separate because this 18-task suite is not an official LIBERO-PRO benchmark.

C. Success-compute trade-off

Fresh re-query invokes the world model 32,883 times on the official protocol. The 50-action latch reduces this to 6,761 checks, a 79.4% reduction, while recording three additional successes. Of those checks, 2,940 world-model proposals are accepted and 3,821 are rejected; the latch bypasses 27,251 otherwise eligible checks. As shown in Fig. 5, the selected controller therefore retains the fused policy’s success level while reducing successful-episode time from 14.11 to 7.35 s. It remains slower than standalone SimVLA (2.69 s), but faster than the standalone world model (14.16 s).

D. Risk-monitor transfer across VLA backbones

Table III summarizes the completed online VLA-only ablations. These rows test portability, not a shared leaderboard: the backbones, feature availability, controller rules and episode manifests differ. The SimVLA six-suite replay gains 11 successes under risk-guided candidate selection. OpenVLA

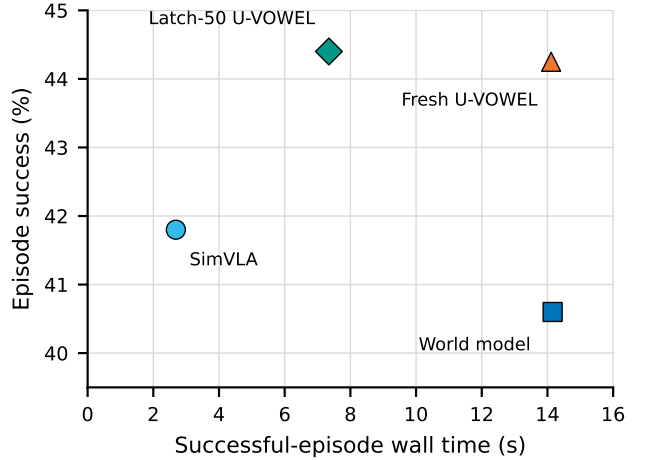


Fig. 5. Official LIBERO-PRO success versus measured wall time on successful episodes. The latch improves the compute profile of fresh routing without a measurable success difference on the matched manifest.

gains 38 successes over H8 when risk controls execution horizon, but its fixed-H1 control reaches 1,022/1,800 (56.78%) and therefore outperforms the adaptive policy. Pi0.5 gains five episodes on Goal-Swap yet loses 18 on OOD18, where the base policy already succeeds on 97.44% of episodes. These mixed outcomes show that offline warning quality does not by itself guarantee beneficial online intervention.

TABLE III

COMPLETED CROSS-BACKBONE ONLINE ABLATIONS. RESULTS ARE INTERNALLY MATCHED WITHIN EACH ROW BUT NOT COMPARABLE ACROSS ROWS.

Backbone and protocol	Reference policy	Risk-aware policy	Successes	Change
SimVLA, six-suite replay	Original H10	Candidate argmin-cap H10	392/680 $\rightarrow$ 403/680	+11
OpenVLA, Goal-Object-OOD	Fixed H8	Adaptive H1/H8	976/1,800 $\rightarrow$ 1,014/1,800	+38
Pi0.5, official Goal-Swap	Basic H10	Selected-cap H10	161/500 $\rightarrow$ 166/500	+5
Pi0.5, official OOD18	Basic H10	Selected-cap H10	1,754/1,800 $\rightarrow$ 1,736/1,800	-18

VII. DISCUSSION

A. Policy complementarity under distribution shift

- The discussion will relate gains to matched episodes solved by only one branch, distinguishing genuine complementarity from changes in execution horizon or query frequency.
- Suite- and task-level patterns will be used to explain which semantic, object and control shifts favor direct action generation or predictive visual reasoning.
- Cases where routing fails will be separated into late alarms, false alarms, rejected useful proposals, prediction errors, action-decoder errors and tasks outside the capability of both branches.

B. Uncertainty as a control signal

- Offline ranking quality will be interpreted through the online decisions it enables. A higher AUROC is relevant only when the selected threshold produces timely interventions at an acceptable false-alarm and compute cost.
- The usefulness of individual uncertainty families will be discussed without assigning a stronger epistemic interpretation than their training objective supports.
- Threshold sensitivity and transfer across task, simulator and hardware conditions will define the practical calibration limits of the current system.

C. Efficiency and deployment implications

- The success–compute frontier will show when selective predictive inference preserves a useful portion of VLA speed and when repeated escalation removes that advantage.
- Query-level and successful-episode timing will be used together to avoid attributing latency differences to policies that simply terminate earlier or fail more often.
- The controller will be positioned as a way to operate complementary learned policies under a budget, not as a replacement for missing policy capability or independent robot safety mechanisms.

D. Limitations

- Episode-level monitor labels do not identify the exact action that makes a trajectory unrecoverable and can reward alarms that occur after the decisive failure.
- Heteroscedastic variance heads model conditional residual dispersion and do not provide a complete estimate of model uncertainty.

- Thresholds are distribution dependent. Their reliability under new tasks, embodiments and hardware conditions must be measured rather than assumed.
- Published state-of-the-art results are not directly comparable when observations, checkpoints, action horizons or episode manifests differ. Matched local reruns remain the primary evidence.
- IsaacLab and real-robot results are limited by available object families, trial counts and safety constraints. Routing cannot recover a skill absent from both policies.

VIII. CONCLUSION

The conclusion will return to the paper’s central question: whether uncertainty can make a fast VLA and a predictive video-world-model policy more effective as a selective system than as standalone policies. It will summarize the final evidence in three parts: the quality and timing of branch-specific failure signals, the success–compute behavior of online fusion and the transfer of policy complementarity from LIBERO and IsaacLab to the physical robot. Exact claims will be limited to completed matched evaluations.

The final paragraph will identify the remaining research directions suggested by the experiments: stronger supervision for warning time, uncertainty that better captures model ignorance, calibration across changing tasks and embodiments and recovery policies that add capabilities beyond those already present in either branch.

REFERENCES

- [1] S. Bai, W. Song, J. Chen, Y. Ji, Z. Zhong, J. Yang, H. Zhao, W. Zhou, W. Zhao, Z. Li, P. Ding, C. Chi, H. Li, C. Xu, X. Zheng, D. Wang, S. Zhang, and B. Chen, “Towards a unified understanding of robot manipulation: A comprehensive survey,” 2025. [Online]. Available: <https://arxiv.org/abs/2510.10903>
- [2] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, G. Gordon, D. Dunson, and M. Dudík, Eds., vol. 15. Fort Lauderdale, FL, USA: PMLR, 11–13 Apr 2011, pp. 627–635. [Online]. Available: <https://proceedings.mlr.press/v15/ross11a.html>
- [3] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, “Vision-language-action models for robotics: A review towards real-world applications,” *IEEE Access*, vol. 13, pp. 162 467–162 504, 2025.
- [4] Y. Luo, W. Chen, T. Liang, B. Wang, and Z. Li, “Simvla: A simple vla baseline for robotic manipulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.18224>
- [5] X. Zhou, Y. Xu, G. Tie, Y. Chen, G. Zhang, D. Chu, P. Zhou, and L. Sun, “Libero-pro: Towards robust and fair evaluation of vision-language-action models beyond memorization,” 2026. [Online]. Available: <https://arxiv.org/abs/2510.03827>
- [6] T. P. Zollo and R. Zemel, “Confidence calibration in vision-language-action models,” 2025. [Online]. Available: <https://arxiv.org/abs/2507.17383>

- [7] Y. Tang, T. Wang, Y. Chen, B. Zhang, Q. Guan, and R. Tang, "Shifting uncertainty to critical moments: Towards reliable uncertainty quantification for vla model," 2026. [Online]. Available: <https://arxiv.org/abs/2603.18342>
- [8] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan, "RoboDreamer: Learning compositional world models for robot imagination," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, and F. Berkenkamp, Eds., vol. 235. PMLR, 21–27 Jul 2024, pp. 61 885–61 896. [Online]. Available: <https://proceedings.mlr.press/v235/zhou24f.html>
- [9] C. Zhu, R. Yu, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, "Unified World Models: Coupling Video and Action Diffusion for Pretraining on Large Robotic Datasets," in *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, June 2025.
- [10] M. J. Kim, Y. Gao, T.-Y. Lin, Y.-C. Lin, Y. Ge, G. Lam, P. Liang, S. Song, M.-Y. Liu, C. Finn, and J. Gu, "Cosmos policy: Fine-tuning video models for visuomotor control and planning," in *The Fourteenth International Conference on Learning Representations*, 2026. [Online]. Available: <https://openreview.net/forum?id=wPEISthXyH>
- [11] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023.
- [12] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. Burchfiel, and S. Song, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023.
- [13] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "Libero: Benchmarking knowledge transfer for lifelong robot learning," in *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., vol. 36. Curran Associates, Inc., 2023, pp. 44 776–44 791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/file/8c3c666820ea055a77726d66fc7d447f-Paper-Datasets.and_Benchmarks.pdf
- [14] A. Kendall and Y. Gal, "What uncertainties do we need in bayesian deep learning for computer vision?" in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/2650d6089a6d640c5e85b2b88265dc2b-Paper.pdf
- [15] M. Seitzer, A. Tavakoli, D. Antic, and G. Martius, "On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=aPOpXlnV1T>
- [16] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/9ef2ed4b7fd2c810847ffa5fa85bce38-Paper.pdf
- [17] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. A. Ambrus, H. Nishimura, and M. Itkina, "Can We Detect Failures Without Failure Data? Uncertainty-Aware Runtime Failure Detection for Imitation Learning Policies," in *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, June 2025.
- [18] R. Römer, A. Kobras, L. Worbis, and A. Schoellig, "Failure prediction at runtime for generative robot policies," in *Advances in Neural Information Processing Systems*, D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, Eds., vol. 38. Curran Associates, Inc., 2025, pp. 7631–7670. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/0b7cb3b8cc44e652761245537027db44-Paper-Conference.pdf
- [19] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and F. Shkurti, "Safe: Multitask failure detection for vision-language-action models," in *Advances in Neural Information Processing Systems*, D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, Eds., vol. 38. Curran Associates, Inc., 2025, pp. 40 041–40 076. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/392d0d05e2f514063e6ce6f8b370834c-Paper-Conference.pdf
- [20] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, p. 494–591, Mar. 2023. [Online]. Available: <https://doi.org/10.1561/22000000101>
- [21] E. J. Hu, yelong shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>
- [22] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=PqvMRDCJT9t>